Start Here

SUBStart Here

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Meeting folder

Everything needed to discuss this work, in one place.

Read in this order

Everything is a PDF. The `.md` files beside them are the editable sources; run `python make_pdfs.py` to rebuild.

#	file	what it is	time
0	`00-plain-english.pdf`	The whole thing with no jargon, plus a glossary	6 min
1	`02-findings.pdf`	The findings, with the numbers	12 min
2	`03-the-ask.pdf`	The one decision I need from you	2 min
3	`papers/01-mittelstadt-2023-*.pdf`, section 6 only	The closest competing work	15 min
4	`04-paper-draft.pdf`	The full paper draft, in academic style	40 min
5	`05-reading-notes.pdf`	What I checked in each cited paper, and what changed	10 min

`01-start-here.pdf` is this document. **If any wording is unfamiliar, start at 00** — it defines every term the others use.

In one sentence

> These fairness fixes make two groups equal either by approving more from the group being turned down, or by rejecting more from the group getting through — and the certifying score cannot tell which happened. Theory published in March 2026 proves it can go either way. **I measured which way, on 26 runs across 15 populations: it depends on how generous the system already is.** And the domains these tools are deployed in sit at opposite ends of that scale — CV screening approves 2–3%, mortgage lending approves 84% — so two organisations can apply the identical tool in good faith and get opposite effects, with identical fairness reports. Their conditions hold on none of my datasets; my rule needs only a historical approval rate, and the two agree at 0.93.

The papers, and why each is here

file	why it matters
01 Mittelstadt et al. (2023)	**The closest competing work.** Showed these fixes mostly harm rather than help, *and* proposed the remedy I thought was mine. Read section 6.
02 Corbett-Davies et al. (2017)	Proves what the mathematically best possible fair program looks like — which makes one of my comparisons a ceiling rather than a rival.
03 Diana et al. (2021)	The alternative method the field recommends for this problem. I test it and it performs badly.
04 Kearns et al. (2018)	Showed fixes can look fine per group while failing for combinations of groups. I reproduce this.
13 Maheshwari et al. (2023)	The other half of paper 04. Showed that these fixes harm combined groups *more*, and that it "often goes unnoticed in the overall performance". My intersectional finding replicates the two of them together; what is mine is ten populations and the condition that it needs a sizeable minority to appear.
11 Backfire (2026)	**The closest work to my main claim.** Proves the effect can go either way when the model cannot see the group — my exact setting. No experiments in it at all. Read the abstract and Theorem 3.
05 Goethals et al. (2024)	**The near miss.** Studies the same dimension I do, calls it "overlooked" — and two of its four authors also wrote paper 01. Their setup fixes the number of approvals in advance, so my effect cannot appear in it.
06 Menon & Williamson (2018)	Supporting theory for paper 02.
07 Ustun et al. (2019)	Another "avoid harm" method from the literature.
08 Black et al. (2022)	On models that score equally well but disagree about individuals. Supports one of my cautions.
09 Agarwal et al. (2018)	The method everything here is built on.
10 Ding et al. (2021)	Argues the field over-relies on one old dataset, and supplies replacements. I use them.

Where the underlying work is

- `../docs/11–26` — 16 documents, one per finding, each stating its own limits
- `../../docs/01–10` — the coursework this grew out of
- `../results/` — every number as a spreadsheet
- `../../tests/test_documented_claims.py` — regenerates every headline figure and fails if a document disagrees with its data
- `git log` — the full record, including predictions committed before the experiments that tested them

Three things I would rather say than be asked

1. My remedy is not new. I believed the fix I proposed was original. It is a variation on one already published (paper 01, section 6). I found this myself by reading their paper, corrected my write-up, and demoted it to a supporting result. What survives: mine works without the program ever seeing which group someone belongs to, and it is tested far more widely.

2. My explanation failed. I tried to derive *why* the flip happens, wrote the prediction down in advance, and tested it on new data. It passed my own test and was then beaten by a much cruder rule, so it explains nothing. It is written up as a failure. I can say when the flip happens, not why.

3. The conditionality itself was published five months before I got to it. A March 2026 theory paper (paper 11) proves the effect can go either way in exactly my setting. I found it by running a novelty check, not before starting. What survives is the empirical half: they have no experiments, their conditions hold on **none** of my 26 runs because the quantity they use diverges on real data, and my rule needs only an approval rate. The two agree at 0.93 — independent theory and independent measurement converging.

Scale

18 independent populations · 326 experimental runs · 1,011 model fits · 2 data sources · 2 kinds of decision · 5 repeats of everything, 12 where the effects were small enough to need them

Four separate attempts to break the finding, on five populations each: a different learner (survived 5/5), a 25× range of constraint strength (4/5), a different route to the same approval rate (4/5), and a different fairness definition (**failed**).